Thesis Research Briefing: Robust Adversarial Defense for Cloud NIDS (V9)

Benchmark: CSE-CIC-IDS2018 (6.32M Clean Flows) | **Evaluation:** 3-Seed Verified (42, 123, 2024) | **Hardware:** AMD Radeon RX 7600 (ROCm / PyTorch)

1. Core Problem Solved & Executive Summary

The Problem Addressed: Deep learning-based Network Intrusion Detection Systems (NIDS) deployed in cloud environments suffer from severe vulnerabilities to adversarial evasion attacks. Conventional defenses exhibit three common limitations: **(1) The Accuracy-Robustness Trade-Off:** Traditional adversarial training often degrades clean network traffic detection. **(2) Overfitting to Seen Attacks:** Training against static attacks (e.g., standard FGSM/PGD) frequently fails when adversaries employ unseen perturbation methods or adaptive step sizes. **(3) Apparent vs. True Robustness:** Defenses may rely on gradient masking or brittle representations rather than enforcing wider margin separations between normal and malicious traffic manifolds.

Headline Summary: *V9 improves clean detection performance and substantially strengthens robustness across multiple bounded white-box, transfer, and gradient-free evaluations. Under unconstrained DeepFool, V9 requires substantially larger perturbations to reach a misclassification, although unrestricted DeepFool can still eventually cross the decision boundary.*

Clean Detection Accuracy	Whitebox PGD Defense	Cross-Attack Defense	DeepFool Perturbation
96.19% +/- 0.03%	**90.13% Acc (6.05% ASR)**	**53% - 74% ASR Drop**	**2.27x Mean L2 Distance**
+2.22% vs Paper Defense (93.97%) Improved clean accuracy	60.3% relative ASR reduction vs Paper Defense (15.23% ASR)	88.5% - 94.1% Acc across APGD, MIM, BIM, CW, Transfer	Required L2=4.15 vs 1.83 97.3% flips exceed eps=0.05

2. Methodology Pipeline & Experimental Configuration



Configuration Parameter	Dataset & Timing Specification	Methodological Purpose & Rationale
Dataset Partitioning	Raw Clean: 6,320,003 flows Train: 4,550,401 | Dev: 505,601 Test: 1,264,001 (Strict Isolation)	Industry-standard cloud network flow benchmark; strict train/test isolation, zero feature/SMOTE leakage
Data Rows after SMOTE	**7,191,590 Balanced Rows** (Applied strictly on Core-Train)	Resolves extreme class imbalance across 14 intrusion classes on training pool only
Feature Selection Counts	40 Candidate (MI) → 15 Paper Space | **27 V9 Space**	Selected using SHAP clean/adversarial stability S(f); retains 80% cumulative stability weight
Training Schedule & Time	Phase 1: 10 ep Clean Pretrain Phase 2: 40 ep Online PGD (5 steps) **Training Time: ~69 min / seed**	Pure-GPU dynamic online batch perturbation generation on AMD RX 7600 (Batch Size = 2048)
Threat Model & Test Time	eps = 0.05 (L_inf feature space) **Evaluation Time: ~15 min / seed** Seeds: [42, 123, 2024]	Standardized L_inf feature-space threat model; 11 threat benchmarks evaluated across all 3 seeds

3. Master Comparative Results: Standard & Whitebox Benchmarks

Direct performance comparison across Baseline MLP, Paper Defense (Static Triad Augmentation), and V9 Defense (Mean +/- Std across 3 seeds):

Threat Benchmark	Category	Baseline (MLP)	Paper Defense	V9 Defense (Ours)	Observed Performance & Improvement
Clean Traffic	Standard Detection	92.82% +/- 0.07% (F1: 0.9303)	93.97% +/- 1.09% (F1: 0.9403)	**96.19% +/- 0.03%** **(F1: 0.9620)**	**+2.22% Clean Accuracy Gain** V9 improved clean accuracy while strengthening defense
PGD (100 steps)	Whitebox Multi-Step	73.90% +/- 2.43% ASR: 20.14%	79.70% +/- 5.45% ASR: 15.23%	**90.13% +/- 1.20%** **ASR: 6.05%**	**+10.43% Acc under attack** **60.3% relative ASR reduction**
FGSM (eps=0.05)	Whitebox Single-Step	84.20% +/- 3.30% ASR: 8.91%	84.32% +/- 3.35% ASR: 9.91%	**92.67% +/- 1.56%** **ASR: 3.62%**	**+8.35% Acc under attack** **63.5% relative ASR reduction**
Square-style Attack	Blackbox Score-Based	82.00% +/- 0.89% ASR: 18.08%	85.40% +/- 4.16% ASR: 13.63%	**91.83% +/- 1.07%** **ASR: 5.27%**	**+6.43% Acc under attack** **61.3% relative ASR reduction**
Transfer PGD	Surrogate Transfer	79.23% +/- 3.62% ASR: 14.90%	86.73% +/- 2.71% ASR: 7.89%	**94.13% +/- 1.68%** **ASR: 2.08%**	**+7.40% Acc under attack** **73.6% relative ASR reduction**
Boundary Probe	Decision Diagnostic	10.17% inside eps (Refine: 100%)	8.11% inside eps (Refine: 100%)	**4.56% inside eps** **(Refine: 100%)**	**43.8% fewer boundary transitions** Fewer samples near the decision threshold

4. Extended Cross-Attack Robustness Evaluation

Evaluated across four distinct optimization formulations **never encountered during training**:

Unseen Attack Variant	Optimization Formulation	Baseline ASR	Paper Defense ASR	V9 Defense (Ours)	Observed Performance & Improvement
Auto-PGD (APGD-CE)	AutoAttack adaptive step size + momentum halving	22.78% +/- 1.44% (Acc: 71.33%)	16.57% +/- 3.39% (Acc: 78.45%)	**7.70% +/- 1.89%** **(Acc: 88.55%)**	**53.5% ASR Reduction (+10.1% Acc)** Resists step-size decay & AutoAttack
MIM (MI-FGSM)	Momentum-iterative ascent (mu=1.0, 40 steps)	16.14% +/- 2.10% (Acc: 77.43%)	14.45% +/- 3.51% (Acc: 80.00%)	**5.09% +/- 1.77%** **(Acc: 91.05%)**	**64.8% ASR Reduction (+11.1% Acc)** Resists velocity-driven gradient directions
BIM (I-FGSM)	Clean-start basic iterative gradient ascent (40 steps)	18.15% +/- 2.81% (Acc: 75.60%)	14.57% +/- 3.30% (Acc: 79.88%)	**4.76% +/- 0.25%** **(Acc: 91.37%)**	**67.3% ASR Reduction (+11.5% Acc)** Strong resistance to clean-start iterative updates
CW-L_inf Margin	Carlini-Wagner logit max-margin loss (Z_other - Z_true)	20.05% +/- 2.09% (Acc: 73.98%)	15.16% +/- 3.96% (Acc: 79.77%)	**6.03% +/- 1.02%** **(Acc: 90.15%)**	**60.2% ASR Reduction (+10.4% Acc)** Overcomes margin loss optimization

5. DeepFool Perturbation Budget Analysis

Evaluation Context: DeepFool operates as an **unconstrained minimum-perturbation attack** without an epsilon limit. It iterates until a classification flip occurs. Evaluating DeepFool requires examining the **magnitude of perturbation** needed to reach the boundary rather than raw success rate alone:

DeepFool Metric	Baseline	Paper Defense	V9 Defense (Ours)	Observed Perturbation Magnitude & Interpretation
Unconstrained Flip Rate	74.98%	68.35%	94.03%	Unbounded search iterates until a boundary crossing is found
Mean L2 Distance to Flip	2.0395	1.8310	**4.1502**	**DeepFool required a 2.27x larger mean L2 perturbation** to find misclassifications
Mean L_inf Needed to Flip	1.1835	1.0748	**2.3238**	Average distortion was 46.5x larger than the nominal eps=0.05 budget
Median L_inf Distortion	0.6141	0.5400	**1.0064**	Approximately 2x higher median distortion than Paper Defense
Flips Exceeding eps=0.05	94.42%	89.77%	**97.33%**	**97.33% of successful DeepFool flips required L_inf > 0.05**
Valid Within-eps (<= 0.05) ASR	5.58%	10.23%	**2.67%**	**Only 2.67% of samples were flipped within the nominal threat budget**

6. Diagnostic Sweeps & Thesis Defensibility

• **Epsilon Decay Curve:** Accuracy degrades from 94.47% (eps=0.01) → 93.35% (0.03) → 90.13% (0.05) → 75.25% (0.10), **showing a smooth degradation in performance as the perturbation budget increases, without an abrupt collapse at the tested budgets**.
• **PGD Step Scaling:** Increasing PGD iterations from 20 to 200 steps maintains 90.13% ± 1.20% accuracy.
• **Evaluation Against Gradient Masking:** The combination of stronger white-box, transfer, and label-only evaluations provides **no simple evidence that the observed robustness is solely an artifact of a weak single-step or under-optimized PGD attack**.

7. Limitations & Scope

1. **Feature-Space Evaluation:** Adversarial perturbations are evaluated under a standardized feature-space L_inf threat model; the functional validity and network-protocol reconstructibility of perturbed flows were not explicitly re-simulated in hardware traffic generators.
2. **Square-style Implementation:** The black-box score-based attack is a custom score-based approximation based on coordinate search principles rather than the exact canonical Square Attack implementation.
3. **Unconstrained Nature of DeepFool:** DeepFool operates as an unconstrained geometric probe and is therefore not directly comparable on raw ASR to bounded L_inf attacks without considering its perturbation distance distribution.

8. Accompanying Thesis Artifacts

Artifact Name	Location	Contents & Description
master_result.csv	Project Root / outputs	232 rows x 83 columns (Complete per-seed runs, sweeps, metadata, and aggregates)
deepfoolbudget.csv	Project Root / outputs	12 rows x 21 columns (Full DeepFool Linf/L2 perturbation distribution analysis)
adversarial_cloud_ids_defense_v9_final.ipynb	Project Root	24 code cells (Fully reproducible end-to-end executable notebook)